Soup | AI Engineer experiment

DON'T SHIP for automatic execution. Run: brev-20261110-main-01. Generated: 2026-11-10T15:30:47.621+00:00.

Choice and protocol

Qwen3-1.7B provides an ungated Apache-2.0 dense Transformer and a manageable T4 footprint. xLAM supplies a concrete function-calling task with independently checkable syntax, function selection, and typed arguments. Source: Salesforce APIGen/xLAM-60k (CC BY 4.0); source-card limitations remain documented.

1000 train / 64 validation / 128 test; naming-based API families separated, normalized prompts deduplicated, cross-split near duplicates excluded. One call, at most 4 tools; literal arguments/defaults only. No truncation, length limit 1024. LoRA r=8, alpha=16, q/v projections; one epoch, lr=0.0002, batch 1, accumulation 8, FP16 compute and FP32 adapter/Adam states. Explicit response+EOT supervision; thinking disabled.

PyTorch/Transformers/PEFT exposes masks, gradients, and effective updates directly; Soup is separately audited. The alternative stack is an observability choice, not evidence of a Soup defect.

Measured artifact comparison

Artifact	Answer NLL	Exact call	JSON valid	Schema valid	Probe failures
base	0.3996	14.8%	32.8%	16.4%	16/48
merged	0.0240	95.3%	100.0%	100.0%	48/48
quantized	0.0338	95.3%	98.4%	98.4%	48/48

4-bit minus base exact-call change: +0.805; paired 95% family-bootstrap interval [+0.670, +0.905], 63 groups. NLL is response-token weighted; its separate paired intervals use per-example NLL. Exact call permits documented optional-default equivalence; raw strict exact is retained.

Memory budget made before training

Hardware deviation: Brev T4 fallback: free Colab initially had no GPU; a later GPU allocation succeeded, but notebook edits, execution, and inspection repeatedly timed out before setup could be verified. Colab results are unclaimed. Brev host RAM differs; measured RSS is reported. All model measurements below come from the declared fallback, not Colab.

GPU: Tesla T4, 14.56 GiB; host RAM: 25.44 GiB. P=1,720,574,976 unique parameters; frozen FP16 weights 2P=3.20 GiB. A=1,605,632; adapter+gradient+Adam moments = 16A bytes. Logits/loss/backward allowance = BSV(2+4+4) bytes, with B=1. Checkpoint activations and a 1 GiB workspace/context allowance are added.

Stage	Estimate GiB	Peak allocated	Peak reserved	Host RSS highwater
train	5.93	4.97	6.05	3.97
merge	8.62	8.45	8.76	3.88
quantize	6.62	1.74	2.26	2.45

Measured allocated minus estimate: train: -0.96 GiB; merge: -0.17 GiB; quantize: -4.88 GiB. Merge estimate: 4P + largest-tensor overlap + per-projection temporary work + allowance. NF4 storage counts packed linear weights, scales, and unquantized embeddings/norms. GPU telemetry is sampled each second and can miss spikes; allocator peaks omit external CUDA memory. RSS includes imports and file-backed pages. Full component arithmetic, stage deltas, and raw device output are in the evidence archive.

What proves training, and what the checks miss

Training effect established: **True**; task improvement established: **True**; export faithful: **True**. 125 optimizer steps were recorded. The verifier checks effective LoRA BA updates, not merely nonzero randomly initialized A tensors.

Enabled/disabled logit RMS difference: 2.35226; repeat-run RMS noise: 0. Active/merged maximum FP32 logit error: 8.4877e-05. Eight validation conversations probe the first/last supervised positions over the full vocabulary. A fresh adapter and a real zero-learning-rate optimizer run are no-op controls. The untouched base as a merged artifact and FP weights as a quantized artifact must be rejected.

Silent failures: evidence and coverage

Failure	Our evidence/check	Soup evidence
Wrong masks / truncation	Prefix equality, shifted labels, EOT and no-truncation assertions; synthetic mask tests.	Data-doctor report and deliberate 32-token control retained; its collator is not ours.
Run completes without learning	Zero-LR control, initial/final BA norms, active/disabled logits.	Live untrained-adapter ship control; inspect its explicit failed rule, not exit code alone.
Adapter lost during merge	Reloaded merged logits vs active adapter; base-as-merged negative control.	Offline ship replays scores and does not itself prove merge parity.
Quantization absent or damaging	Packed NF4 inventory, fresh reload, quality deltas; FP-as-4-bit control.	Offline ship applies supplied task/general scores; no independent artifact inspection.
JSON looks correct but call is wrong	Typed complete-call scoring, wrong-argument tests, raw outputs.	Our scorer tests are independent; Soup general-suite scorer/version and outputs retained.

Verdict and changes required

- 4-bit exact call accuracy is below the committed minimum

- general benchmark regressed: mini_safety

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- behavioral probe produced a non-abstaining or invalid response

- automatic execution is not validated in a representative deployment; sandbox/permission boundaries and larger independent behavioral coverage are required

Before automatic release: resolve failed integrity/quality gates, verify references, expand independent API and adversarial coverage, and validate tool permissions, abstention, and side effects in a representative sandbox. Do not lower thresholds after seeing test results.

Limits and AI use

Public-data holdouts cannot rule out prior training exposure. Naming-based groups can miss API aliases; exact references can reject equivalent answers. The 48 templated probes are correlated. No real external tool was executed. Successful offline tests alone do not establish automatic-execution safety.

AI assisted source comparison, implementation, tests, and report drafting. The initial Qwen2.5/Dolly proposal and generic exact-match scoring were revised after inspection. Concrete checks, corrections, failures, and applicant review status are recorded in DECISIONS.md and EXPERIMENT_LOG.md.